

Nitrogen and oxygen co-doped graphene quantum dots with high capacitance performance for micro-supercapacitors

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ABSTRACT

The applications of carbon-based micro-supercapacitors (MSCs) based on the electrical double layer capacitance mechanism are usually limited by the extremely low specific capacitances and energy storage densities of carbon electrodes fabricated from less active, large-size carbon materials. As a promising alternative, high-activity N and O co-doped graphene quantum dots (N-O-GQDs) offer a combination of advantages, such as ultrasmall sizes, rich active sites, high hydrophilicity, and facile assembly into conductive carbon films. Here we report the facile electrophoresis construction of carbon-based MSCs for ultrahigh energy density storage using N-O-GQDs as the initial electrode material. The N-O-GQD MSCs show extremely high volumetric capacitances of 325 F cm⁻³ in H₂SO₄ due to their high pseudocapacitive activity, high loading density, and enhanced electrolyte wetting ability ascribed to a large amount of doped nitrogen and oxygen functional groups. They deliver an ultrahigh volumetric energy density, superior to that of thin-film lithium batteries. Three connected all-solid-state N-O-GQD MSCs can light a red light-emitting diode. Furthermore, the pseudocapacitive MSCs maintain high power densities, and cycling stability owing to improvements in electrical conductivity and electrolyte penetration. The important results highlight the promising applications of high-activity nanographenes in on-chip power sources for driving diverse micro-devices.

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1. Introduction

The rapid development of portable electronic equipment, implantable medical devices, wireless sensor networks and other micro-electromechanical systems has stimulated an increasing demand for micro-energy storage devices [1–4]. Micro-supercapacitors (MSCs) with electrode sizes of tens to hundreds of micrometers are promising energy storage systems for miniaturized devices due to their excellent rate capability, high power density, and long cycle life [5–9]. Usually, carbon-based MSCs are fabricated based on the electrical double layer capacitance (EDLC) storage mechanism using large-size carbon materials as electrode materials, including carbon fibers (CF) [9], activated carbon (AC) [10], carbide derived carbon (CDC) [2,11], carbon nanotubes (CNT) [7], and graphene sheets [4–6,8]. These carbon-based MSCs

possess light weight, high power density, and excellent cycle life, but their practical applications are limited by low volumetric energy densities (0.6–9 mWh cm⁻³) due to the EDLC storage limitations and the low packing density of these carbon materials with large specific area. In this context, metal oxides (RuO₂ [12] and MnO₂ [13]) and conductive polymers (polyaniline [14] and polypyrrole [15]) are extensively explored as pseudocapacitive materials to enhance the energy densities of MSCs. For example, MSCs based on polyaniline networks and nanofibers were reported to show volumetric energy densities of 5.83 and 16.4 mWh cm⁻³, respectively [16,17]. However, the miniaturization applications of pseudocapacitive MSCs are limited by poor cycle lifetimes and low power densities owing to the low electrical conductivity and sluggish Faradic reaction dynamics of pseudocapacitive materials. The limited enhancement in volumetric energy densities is compromised by poor cycle lifetimes and low power densities, presenting a great challenge in the fabrication of MSCs with comprehensive superior electrochemical performances.

In recent years, ultrasmall nanographenes called graphene

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quantum dots (GQDs) with lateral sizes of several nanometers have attracted increasing attention in electrochemical, photoelectronic, biomedical, and other fields owing to their unique structure and superior properties [18–31]. With rapid development towards device miniaturization, GQDs are more suitable than large carbon materials for the design and fabrication of micro-devices and micro-powers because of their ultrasmall sizes. Moreover, ultrasmall GQDs have much higher edge/core atomic ratios than usual carbon materials, which could greatly enhance electrochemical activities of carbon electrodes, given that edge carbon atoms are typical electrochemically active defects in sp^2 hybrid carbon materials. Many studies have also revealed that the electrochemical activities of carbon materials can be further increased by introducing N or/and O heteroatoms [32–37]. For example, doping carbon materials with N or O heteroatoms could induce additional pseudocapacitive activities in carbon-based supercapacitors and thus greatly increase specific capacitance. Moreover, N or O doping is favorable for improving the interface wettability of the carbon electrode for the aqueous electrolyte and thus enhancing Faradic reaction dynamics. Therefore, N/O co-doped GQDs (N-O-GQDs) with a ultrahigh overall doping concentration are expected to offer ultrahigh specific capacitances and energy storage densities for MSCs owing to their more pronounced edge and doping effects compared with usual carbon materials. Because of these advantages, GQD-based electrodes have recently been assembled for fabrication of symmetric GQD//GQD MSCs and asymmetric (GQD// MnO_2 and GQD//PANI) MSCs. However, reported GQD-based MSCs in aqueous electrolyte showed rather low areal capacitance ($0.5\text{--}3.0\text{ mF cm}^{-2}$) and energy densities ($0.07\text{--}0.41\text{ }\mu\text{Wh cm}^{-2}$) [22,24,25], and their all-solid-state MSCs (GQD//PANI) exhibited poorer performances (0.2 mF cm^{-2} ; $0.03\text{ }\mu\text{Wh cm}^{-2}$) [25]. These unexpected performances are largely ascribed to the low doping level of GQDs and their poor loading ability on micro-electrodes.

Here, we report the first application of heavily co-doped N-O-GQDs in MSCs. N-O-GQDs containing 17.8 at% N and 21.3 at% O were synthesized by a modified molecular fusion method and their loading on interdigital finger gold electrodes was made by electrophoresis. Through the N/O enhanced interaction between the N-O-GQDs and the Au surface of electrodes, the GQDs can be readily deposited on the micro-electrodes to form dense carbon films (3.57 g cm^{-3}) containing self-assembled microtubes. Symmetric MSCs based on the N-O-GQD material offer superior performances in both aqueous and solid electrolytes. Their all-solid-state MSCs in PVA/ H_3PO_4 show extremely high volumetric capacitance (56.1 F cm^{-3}) and volumetric energy density (7.8 mWh cm^{-3}), while their areal capacitance is also high (9.99 mF cm^{-2}) and high cycling stability is maintained. The encouraging results make the N-O-GQD material promising for the next generation of high-performance MSCs.

2. Experimental

Materials: gold interdigital finger electrodes ($20\times 10\times 0.635\text{ mm}$) were purchased from Changchun Megaborui Technology Co., Ltd., China. All other chemicals were of analytical grade and commercially available from Shanghai Chemical Reagent Co. Ltd. and used as received without any further purification.

Fabrication of N-O-GQDs: In a typical procedure for synthesis of N-O-GQDs, pyrene (4 g) was nitrated into trinitropyrene in hot HNO_3 (320 ml) at 80°C under refluxing and stirring for 48 h. After cooled to room temperature, the mixture was diluted with deionized (DI) water and filtered through a $0.22\text{ }\mu\text{m}$ microporous membrane to remove the acid. The resultant yellow 1, 3, 6-trinitropyrene was dispersed in DI water (640 mL) by ultrasonication for 2 h. The

suspension (40 mL) was transferred to a poly (tetrafluoroethylene) (Teflon)-lined autoclave and 4 ml hydrazine hydrate was added, then heated at 200°C for 12 h. After cooled to room temperature, the product containing water-soluble N-O-GQDs was filtered through a $0.22\text{ }\mu\text{m}$ microporous membrane to remove insoluble carbon product. N-O-GQDs powder was obtained by further dialysed in a dialysis bag (retained molecular weight: 3500 Da) for 2 days to remove sodium salt and unfused small molecules and dried at 80°C .

Fabrication of N-O-GQD electrodes: N-O-GQD/Au electrodes were prepared by a facile electrodeposition method with a two-electrode setup, where gold interdigital finger electrodes and Pt foil were used as working electrode and counter electrode, respectively. An aqueous precursor solution with N-O-GQDs (4.5 mg mL^{-1}) and deionized water (40 mL) was used as the electrolyte, and the deposition was performed at a constant potential of $2\text{--}4\text{ V}$ for $1\text{--}4\text{ h}$. After the deposition, the electrodes were washed with distilled water and then dried at air, and obtained the N-O-GQD microelectrodes. In order to explore the contribution of N and O groups of the N-O-GQDs to the pseudocapacitive performance, we annealed the N-O-GQD microelectrodes in argon atmosphere to vary the N and O contents. The microelectrodes treated at 400 and 800°C for 1 h were labeled as N-O-GQD-400 and N-O-GQD-800, respectively.

Fabrication of all-solid-state N-O-GQD MSCs: polyvinyl alcohol (PVA)/ H_3PO_4 hydrogel, as an electrolyte, was synthesized by mixing 6 g of PVA (wt.99.9%) powder, 6 g of H_3PO_4 (wt.85%) and DI water (60 mL). The mixture was heated at 85°C under stirring until the solution turned clear. By casting the hydrogel onto the prepared N-O-GQD micro-electrodes and drying naturally, an all solid-state N-O-GQD MSCs was obtained.

Material characterizations: Scanning electron microscopy (JEOL FESEM-6700F, 15 KV) images were obtained on a field emission scanning electron microanalyzer at an acceleration voltage of 5 kV. Transmission electron microscope (TEM) was operated on a Hitachi H7650 transmission electron microscope with CCD imaging system on an acceleration voltage of 120 kV. XRD patterns were obtained with a Rigaku 18 KWD/max-2550 using $\text{Cu K}\alpha$ radiation. FT-IR spectra of dried samples were recorded with a Bio-Rad FTIR spectrometer FTS165. Raman spectra were recorded on a Renishaw in plus laser Raman spectrometer with $\lambda_{\text{exc}} = 785\text{ nm}$. XPS spectra were collected using a Kratos Axis Ultra DLD X-ray photoelectron spectrometer.

Electrochemical measurements: All electrochemical tests, including cyclic voltammetry (CV), galvanostatic charge–discharge (GCD), and electrochemical impedance spectroscopy (EIS), were carried out using an electrochemical working station (CHI660D, Shanghai, China) at room temperature. In a three-electrode system, N-O-GQD/Au, Ag/AgCl electrode and Pt wire were respectively used as work electrode, the reference electrode, and the counter electrode. The CV tests were measured with the potential window from -0.2 V to 0.8 V in a three-electrode system and from 0 V to 1.0 V in a two-electrode system. The EIS measurements were carried out in the frequency range from 0.01 Hz to 1000 kHz with 5 mV ac amplitude.

In a three-electrode system, the capacitance values of electrodes were calculated from the CV data according to the following formula:

$$C_{\text{electrode}} = \frac{\int i dV}{v \Delta V} \quad (1)$$

Where v is the scan rate (V s^{-1}), ΔV is the operating potential window in volts (1.0 V), and i is the current (A).

In a two-electrode system, the capacitance of each device was

calculated from the galvanostatic curves at different current densities using the formula:

$$C_{\text{device}} = i / (-dV/dt) \quad (2)$$

Where i is the current applied (A), and dV/dt is the slope of the discharge curve (in volts per second, $V s^{-1}$). Specific capacitances were calculated based on the area or the volume of the device stack according to the following formulae:

$$\text{Areal capacitance} = C_{\text{device}}/A \quad (3)$$

$$\text{Volumetric stack capacitance} = C_{\text{device}}/V \quad (4)$$

Where A (cm^2) and V (cm^3) are the geometric surface area of the device is 0.252 cm^2 and geometric volume of substrate. The thickness of N-O-GQD film is about $1.78 \mu\text{m}$. The loading mass of N-O-GQDs is about 0.16 mg .

The energy density (E) and power density (P) were calculated by using $E = \frac{1}{2} C (dV)^2$ and $P = E/dt$, where dV is the potential range and dt is the total time of discharge.

3. Results and discussion

N-O-GQD/Au microelectrodes were fabricated as followed. Firstly, colloidal N-O-GQDs were synthesized through a modified hydrothermal molecular fusion process [19,20] in the presence of hydrazine hydrate using 1,3,6-trinitropyrene as the molecular precursor. To prepare N and O co-doped carbon thin films on interdigital finger Au electrodes, an efficient and rapid electrophoresis process was employed using as-prepared N-O-GQD colloidal solution (Fig. 1a) as an electrolyte without adding other electrolytes because N-O-GQDs under basic conditions are highly negatively charged due to N/O co-doping (Fig. 1b), as confirmed by the zeta potential measurement (Fig. S1, -27.8 mV at pH 11). Owing to the unique surface charge feature of the GQDs, a low potential (typically $+3 \text{ V}$) was applied to the Au electrodes for deposition of

the carbon films with required thickness (Fig. 1c). In contrast, in the electrophoresis of less charged GQDs, an additional electrolyte $\text{Mg}(\text{NO}_3)_2$ and higher DC potential of 80 V were required [22,24,25].

The microstructures of the colloidal GQDs before the deposition were characterized by various techniques. For free-standing colloidal N-O-GQDs, they are highly fluorescent and stable in basic solution for several months without precipitate (Fig. 1a). TEM image of the colloidal GQDs (Fig. 2a) shows that they are well-dispersed in aqueous solution and have a relatively narrow later size distribution ranging from 2.4 nm to 4.8 nm . Their high-resolution TEM (HRTEM) image (Fig. 2b) shows that they are highly graphitized, which are further confirmed by X-Ray Diffraction (XRD) and Raman spectroscopy. Their XRD pattern (Fig. S2a) is characterized by a (002) diffraction peak with a layer spacing $3.44 \text{ \AA}$. Their Raman spectrum (Fig. S2b) demonstrates two typical peaks at 1361.74 and 1586.95 cm^{-1} , corresponding to the D and G band with the I_D/I_G ratio of 0.83 . FT-IR spectrum (Fig. S2c) shows that C=C stretching vibration is at 1610 cm^{-1} and N-H stretching vibrations are at 1640 cm^{-1} and 3210 cm^{-1} . Moreover, the peak at 1280 cm^{-1} is ascribed to the vibration of C-O bonds and a vibration of O-H bonds at 3400 cm^{-1} .

The N-O-GQD/Au microelectrode array was characterized by SEM. The array consists of 28 in-plane interdigital Au microelectrodes (14 positive and 14 negative microelectrodes). Each microelectrode is $180 \mu\text{m}$ in width and 10 mm in length, the distance between adjacent microelectrodes is $150 \mu\text{m}$, and the superficial area A of the N-O-GQD microelectrode is 0.252 cm^2 . Fig. 2c and Fig. 2d shows the top and cross-sectional SEM images of the microelectrode array respectively, where the Au electrode surface is uniformly coated by a dense thin carbon film composed of N-O-GQDs with a thickness h of $\sim 1.78 \mu\text{m}$. Such a small thickness is beneficial to electronic transport and ionic diffusion in the film, thus facilitating reversible and rapid Faradic reactions of active materials. The loading mass of N-O-GQDs is about 0.16 mg , and the film density is as high as 3.57 g cm^{-3} . The high mass density is required for the fabrication of MSCs with a high energy density. Further surface morphological observation (Fig. 2e and Fig. 2f)

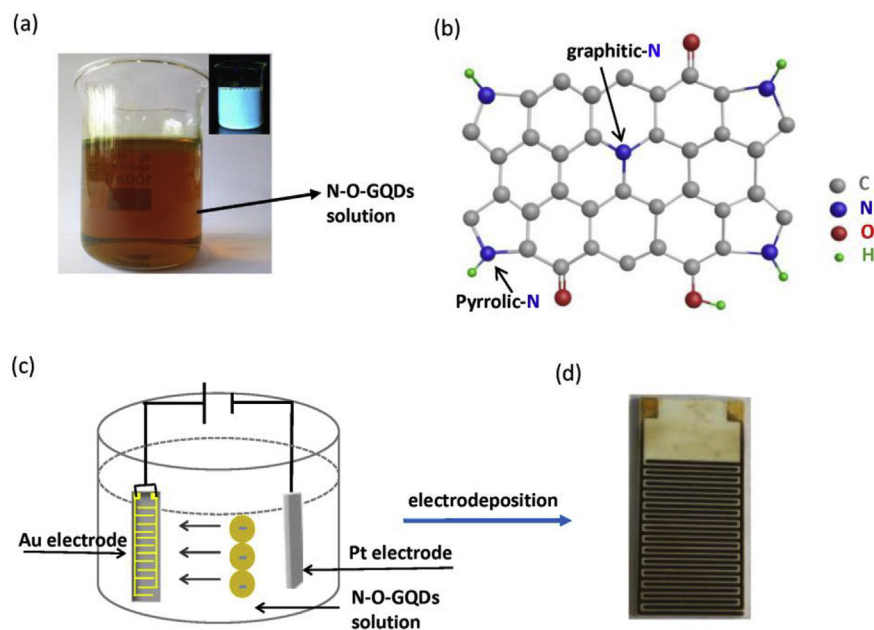


Fig. 1. (a) Photographs of aqueous colloids of N-O-GQDs taken under visible and UV (inset) lights, (b) Schematic model of the functional groups in N-O-GQDs, (c) Schematic illustration of the electrophoretic deposition of N-O-GQDs on the interdigital finger electrode to prepare the N-O-GQD microelectrode, (d) The digital photographs of the N-O-GQD microelectrode. (A colour version of this figure can be viewed online.)

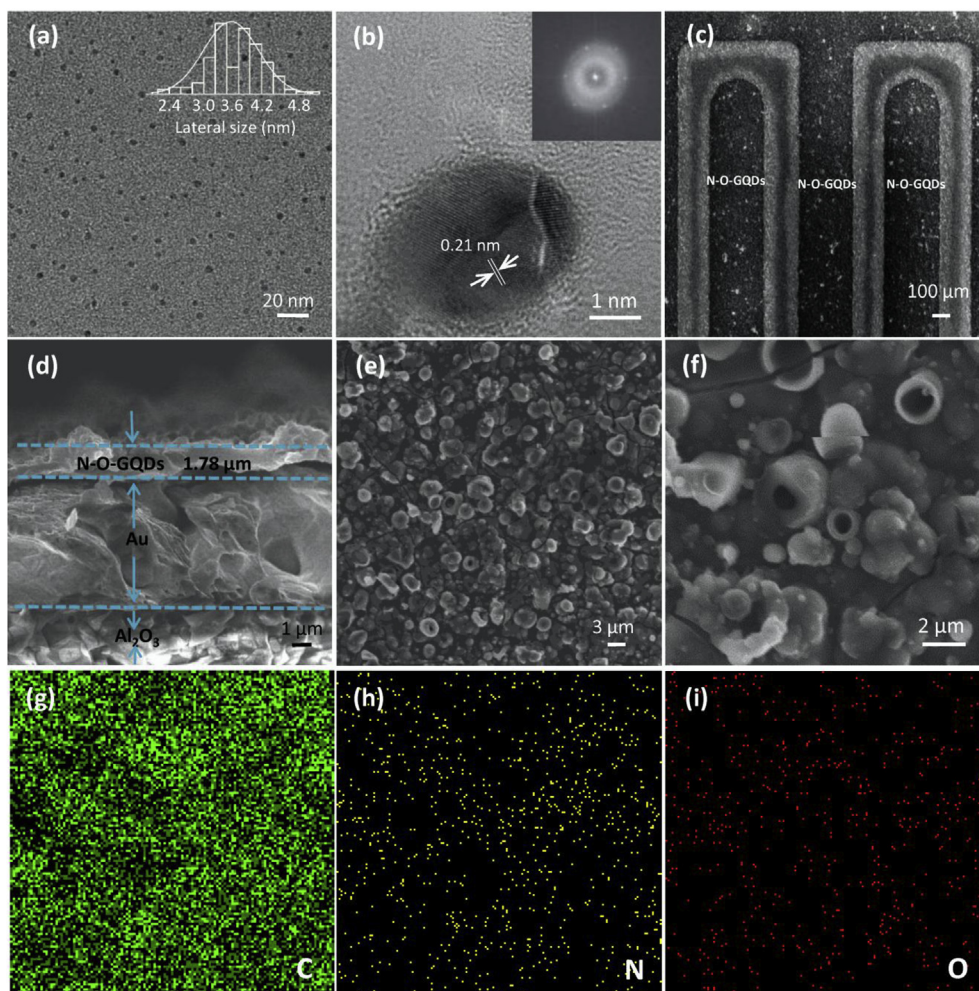


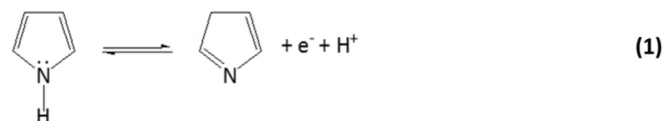
Fig. 2. (a) TEM images of as-prepared N-O-GQDs (insets: lateral size distributions), (b) High-resolution TEM image of an individual N-O-GQD (insets: FFT patterns), (c) SEM images of the interdigital finger Au electrodes after the deposition of N-O-GQD film, (d) Cross-section SEM image of N-O-GQD microelectrode, (e, f) Surface morphologies of N-O-GQD microelectrode, (g) Carbon, (h) Nitrogen, (i) Oxygen EDS map of N-O-GQD microelectrode. (A colour version of this figure can be viewed online.)

shows that a large number of tube-like carbon structures are self-assembled and inserted into the dense film, which facilitates the electrolyte penetration within the interior of the film. The EDS maps of carbon, nitrogen, and oxygen show that the carbon film is homogeneously doped by N and O (Fig. 2 g–i), which is also beneficial to the electrolyte penetration.

The chemical structure of N-O-GQD film were analyzed by X-ray photoelectron spectroscopy (XPS). Its survey XPS spectrum shows strong C 1s, N 1s, and O 1s peaks, and thus the C, N, and O contents are determined to be 60.8 at %, 17.8 at %, and 21.3 at %, respectively (Fig. 3a). The extremely high contents of N and O in the film can greatly enhance the pseudocapacitance activities and surface wettability of the dense and active carbon film, enabling the high-density energy storage for the N-O-GQD MSCs. The high-resolution C 1s spectrum (Fig. 3b) corresponds to the following carbon bonds: 284.6 eV (C=C), 285.5 eV (C-N), 287.2 eV (C-O), and 290.7 eV (C=O). The high-resolution N 1s peak (Fig. 3c) suggests that N dopants exist as pyrrolic N (399.5 eV), graphitic N (401.5 eV) and oxidized N (406.4 eV) [34,38]. The O 1s spectrum (Fig. 3d) shows two separated peaks including 531.2 eV (C=O) and 532.4 eV (O-H).

Dominant pyrrolic N sites are recognized to provide Faradaic pseudocapacitance, whereas low-doped graphitic N is considered as in-graphene nitrogen substitution to improve the electrical conductivity of the carbon film [39,40]. In aqueous acid medium,

pyrrolic N are prone to reversible redox reactions [41], thus offering excellent pseudo-capacitance for MSCs, the possible redox reactions of pyrrolic N (shown in equation (1)) are simply expressed as follows [42].



The O 1s spectrum (Fig. 3d) shows two separated peaks including 531.2 eV (C=O) and 532.4 eV (C-OH). It is known that the acidic functional group reacts with hydroxyl ions (OH^-) in the alkaline aqueous solution and the basic functional group reacts with protons (H^+) in the acidic aqueous solution [43]. Phenolic C-OH is weakly acidic, and it is difficult for it to undergo oxidation/reduction reaction under acidic electrolyte. In contrast, carbonyl groups C=O are susceptible to reversible faradaic redox reactions in 1 M H_2SO_4 electrolyte as following equation: $>\text{C}=\text{O} + \text{H}^+ + e^- \rightleftharpoons >\text{CH}-\text{O}$ (>C represents the carbon network) [41]. The abundant oxygen-containing groups are advantageous, as they can provide a large additional pseudocapacitance as well as improved wettability.

The chemical structures of N-O-GQD-400 and N-O-GQD-800 were analyzed by XPS (Fig. S3 and Fig. S4). For comparison, the N

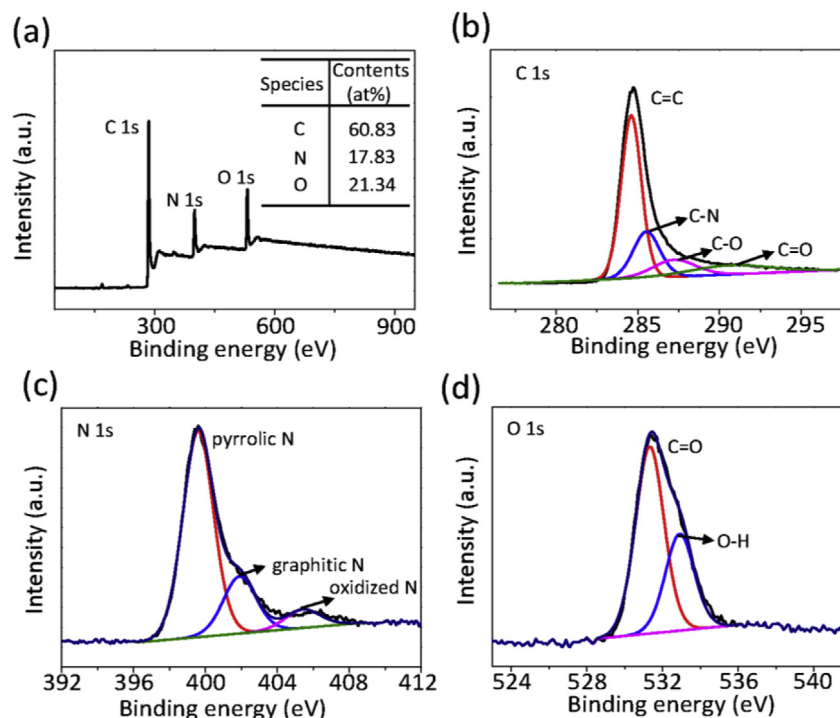


Fig. 3. (a) The wide survey XPS spectrum of N-O-GQD electrode material, (b) C 1s spectrum, (c) N 1s spectrum, (d) O 1s spectrum. (A colour version of this figure can be viewed online.)

content of the unannealed N-O-GQD and annealed N-O-GQD-400 and N-O-GQD-800 samples is listed in Table 1. For the unannealed sample, the electrochemically active N content (pyrrolic N) is as high as 10.35 at %. For 400 °C annealing, however, the total active N content (pyrrolic N and pyridinic N) dramatically decreases to 2.95 at %. For further annealing at 800 °C (N-O-GQD-800), the active N content is reduced to 1.68 at %.

The O content of the three samples is also listed in Table 1. The active C=O content of unannealed N-O-GQDs is 7.63 at % (-OH oxygen is as high as 13.67 at % but not active [43]). After annealing at 400 °C, the active O content (C=O) is nearly unchanged (7.25 at %). After annealing at 800 °C, the C=O content significantly decreases to 3.53 at %.

CV measurements were carried out to analyze the electrochemical behavior of the N-O-GQD, N-O-GQD-400 and N-O-GQD-800 microelectrodes using a three-electrode configuration with the potential ranging from -0.2 to 0.8 V at 10 V s^{-1} in $1.0 \text{ M H}_2\text{SO}_4$ (Fig. 4a). For the unannealed (N-O-GQD) and low-temperature-annealed (N-O-GQD-400) microelectrodes, their CV curves display a typical pseudocapacitive feature. For the high-temperature-annealed microelectrode (N-O-GQD-800), in contrast, the CV curve shows a rectangular shape, implying the typical EDLC characteristic. Also, we measured the volumetric capacitance of the three samples (Fig. 4b). We found that with

increasing annealing temperature, the volumetric capacitance is reduced from 91.4 to 52.6 F cm^{-3} (N-O-GQD-400) and then to 27.0 F cm^{-3} (N-O-GQD-800).

The observation of the annealing-dependent electrochemical behaviors of the N-O-GQD microelectrodes indicates that the both contents and configurations of N and O exert an important influence on their electrochemical performance. For the unannealed sample, the highest pseudocapacitive electrochemical performance is dominated by active pyrrolic N and active C=O groups with rather high contents [44]. For the low-temperature-annealed sample, the reduced volumetric capacitance is largely ascribed to the decrease in the content of the overall active N sites. For the high-temperature-annealed sample, the pseudocapacitive activities are lost owing to the removal of most N and O active sites, leading to the EDLC behavior.

A series of N-O-GQD films electrodeposited on Au microelectrodes were prepared under different DC potentials (1–3 V) and deposition durations (1–4 h). Their capacitive performance was evaluated by cyclic voltammetry measurements at scan rates 10 V s^{-1} in $1 \text{ M H}_2\text{SO}_4$ (Fig. S5). The volumetric capacitance, loading mass, and film thickness (Fig. S6) were measured and listed in Table S1 for the different deposition conditions. We found that the deposition under higher potential for longer durations produced thicker films with larger loading masses (Table S1). The deposition

Table 1

The elemental composition and quantities (at %) of different energy levels atomic ratios for N-O-GQD, N-O-GQD-400 and N-O-GQD-800 microelectrode materials obtained from XPS.

Sample	Element content Form XPS (at %)			Nitrogen content (at %)				Oxygen content (at %)	
	C	N	O	pyridinic N	Pyrrolic N	Graphitic N	Oxidized N	C=O	O-H
N-O-GQDs	60.8	17.8	21.3	—	10.35	6.28	1.16	7.63	13.67
N-O-GQDs-400	86.24	3.62	10.13	2.28	0.67	0.67	—	7.25	2.88
N-O-GQDs-800	92.73	2.33	4.94	0.96	0.72	0.65	—	3.53	1.41

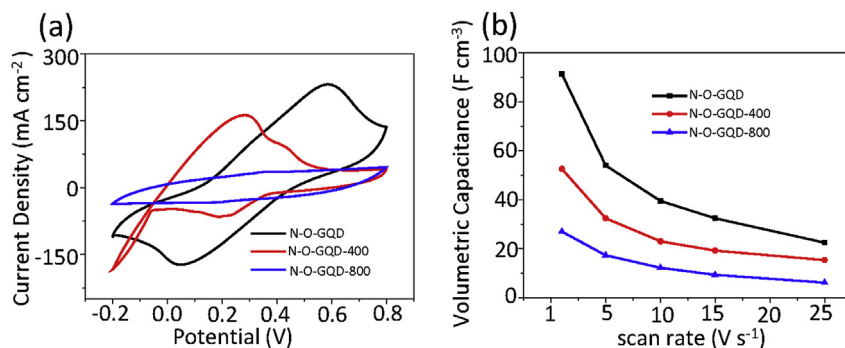


Fig. 4. (a) Cyclic voltammetry curves of N-O-GQD, N-O-GQD-400 and N-O-GQD-800 microelectrodes at 10 V s^{-1} in $1.0 \text{ M H}_2\text{SO}_4$ electrolyte using a three-electrode configuration, (b) Comparison of volumetric capacitance values with increasing scan rates for N-O-GQD, N-O-GQD-400 and N-O-GQD-800 microelectrodes (from 1 to 25 V s^{-1}). (A colour version of this figure can be viewed online.)

condition for optimizing the film thickness, loading mass, and thus volumetric capacitance was determined to be 3 V for 2 h . SEM imaging on the film surface morphology further shows that under the optimized condition ($3 \text{ V } 2 \text{ h}$), some opened nanotubes are formed and inserted in the film, while at shorter deposition duration (1 h), the unique nanotube structure is not formed, but at longer duration (3 h) the formed nanotubes are blocked (Fig. S7). We believe that the opened nanotube structures may facilitate the electrolyte penetration in the thick film, leading to the higher volumetric capacitance.

The CV curves of optimized N-O-GQD MSCs exhibit a large capacitive response, accompanied by relatively wide redox peaks, indicating the presence of a pseudocapacitive effect with N/O co-doping (Fig. 5a). Remarkably, N-O-GQD MSCs allow for operation at an ultrahigh scan rate of up to 500 V s^{-1} , suggesting superior rate capability and excellent electrochemical reversibility [45]. Fig. 5b shows the galvanostatic charge/discharge curves of N-O-GQD MSCs at different current densities. The charge/discharge curves at low current density of $150 \mu\text{A cm}^{-2}$ are unsymmetrical because a redox

reaction generally takes more time than electric double layer formation [46]. Fig. 5c shows the volumetric capacitances at different current densities from 0.8 to 3.9 A cm^{-3} . The N-O-GQD MSC exhibits an ultrahigh volumetric capacitance C_V of 325.5 F cm^{-3} at 0.8 A cm^{-3} and 73.9 F cm^{-3} at 3.9 A cm^{-3} , a high areal capacitance C_A of 57.9 mF cm^{-2} at the current density of $150 \mu\text{A cm}^{-2}$ (Fig. S8a). The areal capacitance of our N-O-GQD MSCs is 57.9 mF cm^{-2} , much higher than the highest areal capacitance (2.98 mF cm^{-2}) of undoped GQD-based MSCs reported so far (Table S2). The comparison indicates that greatly enhanced electrochemical performance of N-O-doped GQD MSCs is largely attributed to the co-existence of active pyrrolic N and carbonyl group with a high concentration as well as the opened carbon nanotubes inserted into the thick film [47–49]. Owing to the ultrahigh areal/volumetric capacitances, a high energy density of 45.2 mWh cm^{-3} ($8.0 \mu\text{Wh cm}^{-2}$) for N-O-GQD MSCs is achieved with the power density of 421 mW cm^{-3} ($75 \mu\text{W cm}^{-2}$) at the current density of 0.8 A cm^{-3} (Fig. 5d, Fig. S8b), much higher than those of lithium thin film batteries ($1\text{--}10 \text{ mWh cm}^{-3}$) [50].

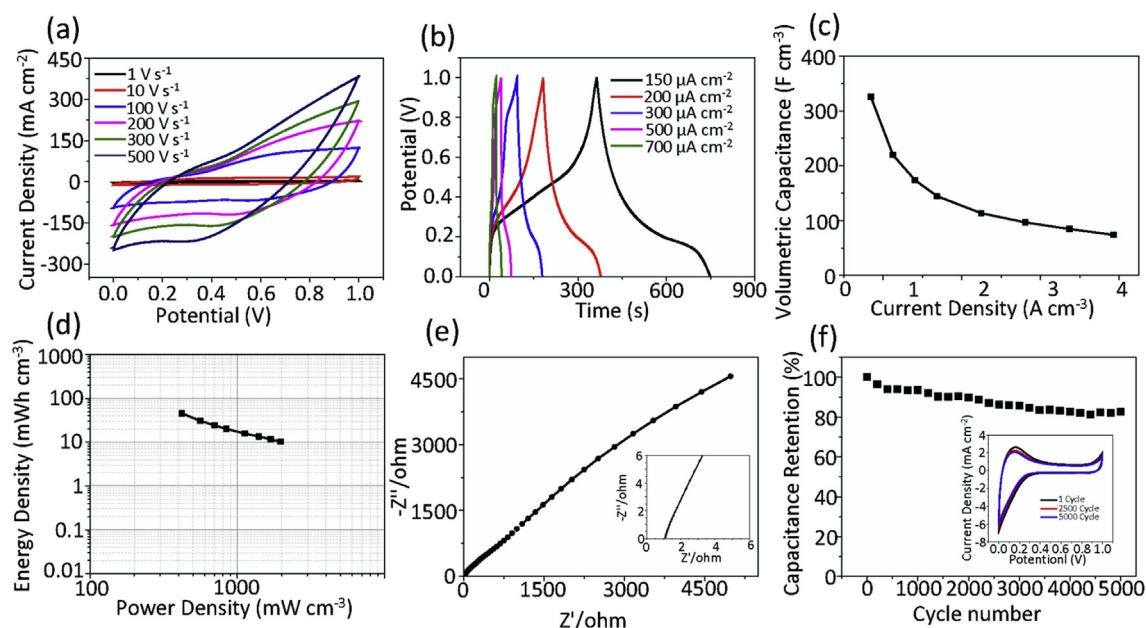


Fig. 5. Two-electrode electrochemical performance of N-O-GQD MSCs in $1 \text{ M H}_2\text{SO}_4$ electrolyte. (a) Cyclic voltammetry curves at various scan rates from 1 to 500 V s^{-1} , (b) Galvanostatic charge/discharge curves at $150\text{--}700 \mu\text{A cm}^{-2}$, (c) The volumetric capacitance at different current densities, (d) Ragone plot of N-O-GQD MSCs showing the relationship of specific volumetric energy density and power density, (e) Nyquist plot of N-O-GQD MSCs, and the inset is an expanded view in the region of high frequencies, (f) Cycling stability at 1 V s^{-1} , and the inset is the CV curves of the 1st, 2500th and 5000th cycles. (A colour version of this figure can be viewed online.)

Nyquist plots (Fig. 5e) were used to understand the high performance of the N-O-GQD MSCs, which provided insights into the impedance distribution and ion diffusion. The semicircle diameter represents the charge transfer resistance (R_{ct}), which is related to the resistances at the interfaces of the active material/current collector and the electrode/electrolyte, no semi-circle appears at the high frequency region for the device (Inset in Fig. 5e), suggesting excellent electrical contact and the fast electron and ion transport/diffusion at the interface between the N-O-GQD film and the gold current collector [51]. The internal or equivalent series resistance (R_s), calculated from the intercept at the Z' -axis in the region of high frequency was 1.1Ω , which indicates further confirmed the better electrical conductivity. The cycling stability of the N-O-GQD MSC is presented in Fig. 5f. The device displays an approximately 82.6% retention of its initial specific capacitance after 5000 continuous cycles at a scan rate of 1 V s^{-1} . The inset shows that there is no obvious change among the 1st, 2500th and 5000th cycles of the N-O-GQD MSC.

The electrochemical performance of all-solid-state N-O-GQD MSCs was also evaluated using PVA/ H_3PO_4 gel as the solid electrolyte. Fig. 6a shows the CV curves of the all-solid-state device at various scan rates ($1\text{--}500 \text{ V s}^{-1}$) with almost alike shapes within $0\text{--}1 \text{ V}$. Fig. 6b shows that the charge-discharge curves of the all-solid-state device at various current densities are not regular triangle and deviate from linearity, which is mainly ascribed to the pseudo-faradaic processes on the electrode surface. The discharging process near 0 V is much slower than the process above $\sim 0.1 \text{ V}$, suggesting that this may be related to sluggish ion diffusion due to the use of a solid electrolyte with a low ionic conductivity [33,41,45,50,52]. The volumetric capacitance C_V of the device is up to 56.1 F cm^{-3} and the areal capacitance is 9.99 mF cm^{-2} at a current density of $50 \mu\text{A cm}^{-2}$ and 6.88 F cm^{-3} and the areal capacitance is 1.23 mF cm^{-2} at a current density of $350 \mu\text{A cm}^{-2}$ (Fig. 6c and Fig. S9a). Furthermore, the all-solid-state MSC exhibits an extremely high volumetric/areal energy density of 7.8 mWh cm^{-3} ($1.4 \mu\text{Wh cm}^{-2}$) at the power density of 140.4 mW cm^{-3} ($25 \mu\text{W cm}^{-2}$) at the current density of 0.28 A cm^{-3} (Fig. 6d,

Fig. S9b), which is about ten times higher than the energy densities of commercially available supercapacitors ($<1 \text{ mWh cm}^{-3}$), and even superior to the thin-film lithium battery ($0.3\text{--}10 \text{ mWh cm}^{-3}$). This value is also higher than that of recently reported all-solid-state MSCs based on graphene materials and other pseudo-capacitive materials, including GQDs/PANI ($0.029 \mu\text{Wh cm}^{-2}$ at $15 \mu\text{A cm}^{-2}$) [25], rGO film (2.5 mWh cm^{-3} at 0.01 V s^{-1}) [50], SWNT/rGO (6.3 mWh cm^{-3} at 26.7 mA cm^{-3}) [52], N-doped rGO (0.3 mWh cm^{-3} at $20 \mu\text{A cm}^{-2}$) [33], PANI (5.83 mWh cm^{-3} at 0.1 mA cm^{-2}) [17] and $\text{Co}(\text{OH})_2/\text{rGO}$ ($0.35 \mu\text{Wh cm}^{-2}$ at 0.05 mA cm^{-2}) [53]. To our knowledge, the volumetric energy density in this work is one of the highest values among solid-state MSCs reported to date (Table S3). As shown in Fig. 6e, there are no semi-circles at the high frequency region, revealing excellent electrical connection at the interface between the thin carbon film and the gold current collector. Meanwhile, R_s is low ($\sim 5.2 \Omega$), indicative of a high electrical conductivity of the N-O-doped carbon film. Furthermore, the straight line of the Nyquist plot represents the Warburg impedance (R_w). Clearly, the line slope is not very precipitous and approach 45° , revealing a bit of slow ion diffusion. This is due to lower ionic conductivity in the solid electrolyte. Fig. 6f shows that the all-solid-state N-O-GQD MSC retains 82.4% of its initial capacity after 5000 cycles under the scan rate of 1 V s^{-1} , indicating its long cycle life. The inset shows that there is no obvious change among the 1st, 2500th and 5000th cycles of the N-O-GQD MSC. The cycling stability of our device is higher than that of other MSCs based on pseudo-capacitive materials, such as nanostructured MnO_2 , WO_3 , PANI, PPy or their composites [16,25,54,55].

In order to produce a reasonable output voltage or output current for practical applications, three all-solid-state N-O-GQD MSCs are assembled in series (Fig. 7) and in parallel (Fig. S10). Fig. 7a shows the CV curves of a single MSC and three N-O-GQD MSCs in series. The potential window is increased from 1.0 V for one unit to 3.0 V for the tandem device. Meanwhile, the tandem device shows almost unchanged charge/discharge time compared with the individual unit at the same current density $100 \mu\text{A cm}^{-2}$ (Fig. 7b). The electrochemical performance of parallel MSCs is also shown

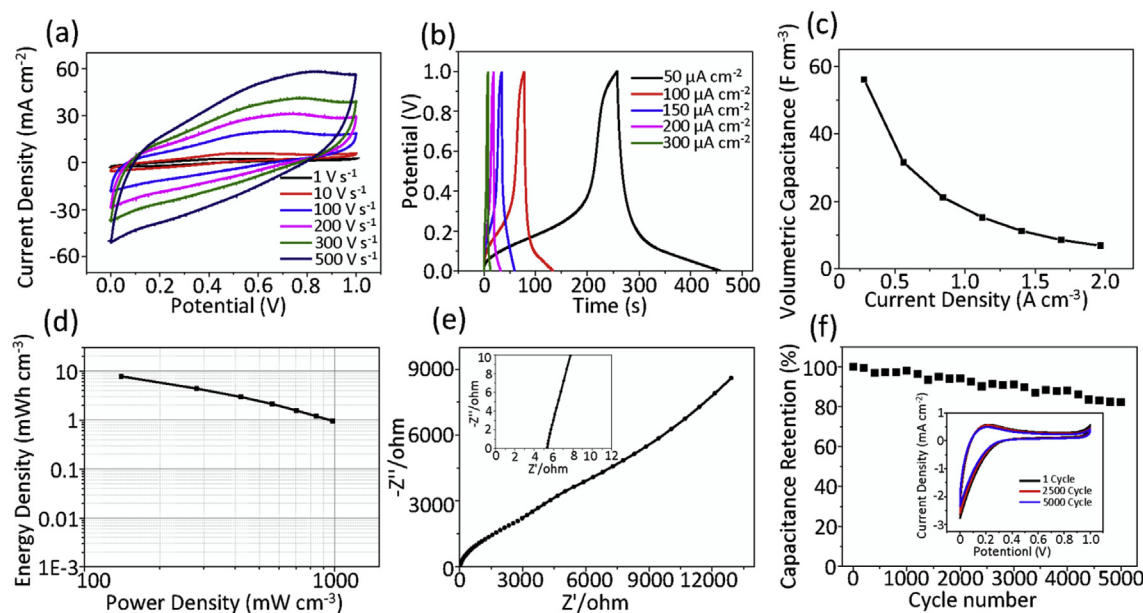


Fig. 6. Two-electrode electrochemical performance of N-O-GQD MSCs in PVA/ H_3PO_4 solid state electrolyte. (a) Cyclic voltammetry curves at various scan rates from 1 to 500 V s^{-1} , (b) Galvanostatic charge/discharge curves at $50\text{--}300 \mu\text{A cm}^{-2}$, (c) The volumetric capacitances at different current densities, (d) Ragone plot of N-O-GQD MSCs showing the relationship of specific volumetric energy density and power density, (e) Nyquist plot of N-O-GQD MSCs, and the inset is an expanded view in the region of high frequencies, (f) Cycling stability at 1 V s^{-1} , and the inset is the CV curves of the 1st, 2500th and 5000th cycles. (A colour version of this figure can be viewed online.)

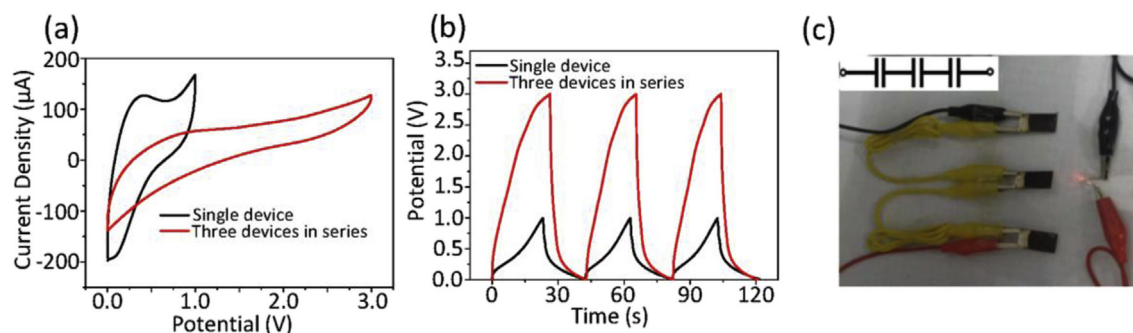


Fig. 7. (a) CV curves of a single device and three devices connected in series at 1 V s^{-1} , (b) Galvanostatic charge/discharge curves of three devices connected in series at $100 \mu\text{A cm}^{-2}$. A single device is shown for comparison, (c) The digital image of a commercial LED lighting up by using three all-solid-state N-O-GQD MSCs devices connected in series. (A colour version of this figure can be viewed online.)

(Fig. S10). The output current of the parallel MSCs is three times than that of a single MSC, and the discharge time is triple. N-O-GQD MSCs can work collaboratively when connected in parallel or in series, which can offer a nano-/micro-scale power source sufficient to meet certain applications that require higher operating currents and voltages. Indeed, a red light-emitting diode is lighted up by three all-solid-state N-O-GQD MSCs in series (Fig. 7c), demonstrating practical applications for powering other electronic devices.

4. Conclusions

In summary, we fabricated N-O-GQD MSCs with a facile electro-deposition method using N/O co-doped GQDs as the active material and H_2SO_4 , PVA/ H_3PO_4 as the electrolyte respectively. The N-O-GQD MSCs show extremely high volumetric capacitance (325 F cm^{-3} in H_2SO_4 , 56.1 F cm^{-3} in PVA/ H_3PO_4) due to enhanced electrolyte-electrode interaction, additional pseudocapacitance, modification polarity of carbon matrices to improve electrolyte wettability through appropriate nitrogen and oxygen doping. The areal/volumetric energy density of this all-solid-state device is up to $1.4 \mu\text{Wh cm}^{-2}/7.8 \text{ mWh cm}^{-3}$, which is one of the highest values among that of reported all-solid-state MSCs based on graphene materials and other pseudocapacitive materials. For practical applications, three all-solid-state N-O-GQD MSCs connected in series can light a red light-emitting diode. Furthermore, the MSCs have good cycling stability. These results demonstrate that N-O-GQD MSCs have great potential to be applied as on-chip power sources for micro-devices.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.carbon.2018.06.042>.

References

- [1] S.E. Moosavifard, J. Shamsi, M.K. Altafi, Z.S. Moosavifard, All-solid state, flexible, high-energy integrated hybrid micro-supercapacitors based on 3D LSG/CoNi₂S₄ nanosheets, *Chem. Commun.* 52 (89) (2016) 13140–13143.
- [2] J. Chmiola, C. Largeot, P.-L. Taberna, P. Simon, Y. Gogotsi, Monolithic carbide-

- derived carbon films for micro-supercapacitors, *Science* 328 (5977) (2010) 480–483.
- [3] D. Pech, M. Brunet, H. Durou, P. Huang, V. Mochalin, Y. Gogotsi, et al., Ultra-high-power micrometre-sized supercapacitors based on onion-like carbon, *Nat. Nanotechnol.* 5 (9) (2010) 651–654.
- [4] S.S. Delekt, A.D. Smith, J. Li, M. Ostling, Inkjet printed highly transparent and flexible graphene micro-supercapacitors, *Nanoscale* 9 (21) (2017) 6998–7005.
- [5] M.F. El-Kady, R.B. Kaner, Scalable fabrication of high-power graphene micro-supercapacitors for flexible and on-chip energy storage, *Nat. Commun.* 4 (2013) 1475–1483.
- [6] M.F. El-Kady, V. Strong, S. Dubin, R.B. Kaner, Laser scribing of high-performance and flexible graphene-based electrochemical capacitors, *Science* 335 (6074) (2012) 1326–1330.
- [7] S.K. Kim, H.J. Koo, A. Lee, P.V. Braun, Selective wetting-induced micro-electrode patterning for flexible micro-supercapacitors, *Adv. Mater.* 26 (30) (2014) 5108–5112.
- [8] J.J. Yoo, K. Balakrishnan, J. Huang, V. Meunier, B.G. Sumpter, A. Srivastava, et al., Ultrathin planar graphene supercapacitors, *Nano Lett.* 11 (4) (2011) 1423–1427.
- [9] J. Bae, M.K. Song, Y.J. Park, J.M. Kim, M. Liu, Z.L. Wang, Fiber supercapacitors made of nanowire-fiber hybrid structures for wearable/flexible energy storage, *Angew. Chem.* 50 (7) (2011) 1683–1687.
- [10] D. Pech, M. Brunet, P.-L. Taberna, P. Simon, N. Fabre, F. Mesnilgrante, et al., Elaboration of a microstructured inkjet-printed carbon electrochemical capacitor, *J. Power Sources* 195 (4) (2010) 1266–1269.
- [11] M. Heon, S. Lofland, J. Applegate, R. Nolte, E. Cortes, J.D. Hettinger, et al., Continuous carbide-derived carbon films with high volumetric capacitance, *Energy Environ. Sci.* 4 (1) (2011) 135–138.
- [12] Z.J. Han, S. Pineda, A.T. Murdock, D.H. Seo, K. Ostrikov, A. Bendavid, RuO₂-coated vertical graphene hybrid electrodes for high-performance solid-state supercapacitors, *J. Mater. Chem. A* 5 (33) (2017) 17293–17301.
- [13] M. Xue, Z. Xie, L. Zhang, X. Ma, X. Wu, Y. Guo, et al., Microfluidic etching for fabrication of flexible and all-solid-state micro-supercapacitor based on MnO₂ nanoparticles, *Nanoscale* 3 (7) (2011) 2703–2708.
- [14] K. Wang, W. Zou, B. Quan, A. Yu, H. Wu, P. Jiang, et al., An all-solid-state flexible micro-supercapacitor on a chip, *Adv. Energy Mater.* 1 (6) (2011) 1068–1072.
- [15] M. Beidaghi, C. Wang, Micro-supercapacitors based on three dimensional interdigital polypyrrole/C-MEMS electrodes, *Electrochim. Acta* 56 (25) (2011) 9508–9514.
- [16] J. Maeng, Y.J. Kim, C. Meng, P.P. Irazoqui, Three-Dimensional microcavity array electrodes for high-capacitance all-solid-state flexible microsupercapacitors, *ACS Appl. Mater. Interfaces* 8 (21) (2016) 13458–13465.
- [17] H. Hu, K. Zhang, S. Li, S. Ji, C. Ye, Flexible, in-plane, and all-solid-state micro-supercapacitors based on printed interdigital Au/polyaniline network hybrid electrodes on a chip, *J. Mater. Chem. A* 2 (48) (2014) 20916–20922.
- [18] Y. Liu, Y. Guo, M. Li, L. Wang, B. Geng, Z. Chen, et al., A bionic strategy for addressing scale-span issues in all-carbon electrocatalytic systems, *Electrochim. Acta* 245 (2017) 310–318.
- [19] D. Pan, J. Zhang, Z. Li, M. Wu, Hydrothermal route for cutting graphene sheets into blue-luminescent graphene quantum dots, *Adv. Mater.* 22 (6) (2010) 734–738.
- [20] L. Wang, Y. Wang, T. Xu, H. Liao, C. Yao, Y. Liu, et al., Gram-scale synthesis of single-crystalline graphene quantum dots with superior optical properties, *Nat. Commun.* 5 (2014) 5357–5365.
- [21] Z. Li, P. Qin, L. Wang, C. Yang, Y. Li, Z. Chen, et al., Amine-enriched graphene quantum dots for high-pseudocapacitance supercapacitors, *Electrochim. Acta* 208 (2016) 260–266.
- [22] B. Shen, J. Lang, R. Guo, X. Zhang, X. Yan, Engineering the electrochemical capacitive properties of microsupercapacitors based on graphene quantum dots/MnO₂ using ionic liquid gel electrolytes, *ACS Appl. Mater. Interfaces* 7 (45) (2015) 25378–25389.

- [23] Q. Xue, H. Huang, L. Wang, Z. Chen, M. Wu, Z. Li, et al., Nearly monodisperse graphene quantum dots fabricated by amine-assisted cutting and ultrafiltration, *Nanoscale* 5 (24) (2013) 12098–12103.
- [24] W.W. Liu, Y.Q. Feng, X.B. Yan, J.T. Chen, Q.J. Xue, Superior micro-supercapacitors based on graphene quantum dots, *Adv. Funct. Mater.* 23 (33) (2013) 4111–4122.
- [25] W.W. Liu, X.B. Yan, J.T. Chen, Y.Q. Feng, Q.J. Xue, Novel and high-performance asymmetric micro-supercapacitors based on graphene quantum dots and polyaniline nanofibers, *Nanoscale* 5 (13) (2013) 6053–6062.
- [26] D. Pan, J. Jiao, Z. Li, Y. Guo, C. Feng, Y. Liu, et al., Efficient separation of electron–hole Pairs in graphene quantum dots by TiO₂ heterojunctions for dye degradation, *ACS Sustain. Chem. Eng.* 3 (10) (2015) 2405–2413.
- [27] J. Zhao, J. Lu, L. Wang, L. Tian, X. Deng, L. Tian, et al., Ultrafast spontaneous emission modulation of graphene quantum dots interacting with Ag nanoparticles in solution, *Appl. Phys. Lett.* 109 (2) (2016), 021905.
- [28] L. Wang, W. Li, B. Wu, Z. Li, S. Wang, Y. Liu, et al., Facile synthesis of fluorescent graphene quantum dots from coffee grounds for bioimaging and sensing, *Chem. Eng. J.* 300 (2016) 75–82.
- [29] D. Pan, L. Guo, J. Zhang, C. Xi, Q. Xue, H. Huang, et al., Cutting sp² clusters in graphene sheets into colloidal graphene quantum dots with strong green fluorescence, *J. Mater. Chem.* 22 (8) (2012) 3314–3318.
- [30] L. Wang, W. Li, B. Wu, Z. Li, D. Pan, M. Wu, Room-temperature synthesis of graphene quantum dots via electron-beam irradiation and their application in cell imaging, *Chem. Eng. J.* 309 (2017) 374–380.
- [31] K. Lee, H. Lee, Y. Shin, Y. Yoon, D. Kim, H. Lee, Highly transparent and flexible supercapacitors using graphene-graphene quantum dots chelate, *Nanomater. Energy* 26 (2016) 746–754.
- [32] T. Lin, I.-W. Chen, F. Liu, C. Yang, H. Bi, F. Xu, et al., Nitrogen-doped mesoporous carbon of extraordinary capacitance for electrochemical energy storage, *Science* 350 (6267) (2015) 1508–1513.
- [33] S. Liu, J. Xie, H. Li, Y. Wang, H.Y. Yang, T. Zhu, et al., Nitrogen-doped reduced graphene oxide for high-performance flexible all-solid-state micro-supercapacitors, *J. Mater. Chem. A* 2 (42) (2014) 18125–18131.
- [34] D. Yu, K. Goh, Q. Zhang, L. Wei, H. Wang, W. Jiang, et al., Controlled functionalization of carbonaceous fibers for asymmetric solid-state micro-supercapacitors with high volumetric energy density, *Adv. Mater.* 26 (39) (2014) 6790–6797.
- [35] H. Oda, A. Yamashita, S. Minoura, M. Okamoto, T. Morimoto, Modification of the oxygen-containing functional group on activated carbon fiber in electrodes of an electric double-layer capacitor, *J. Power Sources* 158 (2) (2006) 1510–1516.
- [36] L.-Z. Fan, S. Qiao, W. Song, M. Wu, X. He, X. Qu, Effects of the functional groups on the electrochemical properties of ordered porous carbon for supercapacitors, *Electrochim. Acta* 105 (2013) 299–304.
- [37] H. Liu, H. Song, X. Chen, S. Zhang, J. Zhou, Z. Ma, Effects of nitrogen- and oxygen-containing functional groups of activated carbon nanotubes on the electrochemical performance in supercapacitors, *J. Power Sources* 285 (2015) 303–309.
- [38] M. Zhou, F. Pu, Z. Wang, S. Guan, Nitrogen-doped porous carbons through KOH activation with superior performance in supercapacitors, *Carbon* 68 (2014) 185–194.
- [39] C. Long, D. Qi, T. Wei, J. Yan, L. Jiang, Z. Fan, Nitrogen-doped carbon networks for high energy density supercapacitors derived from polyaniline coated bacterial cellulose, *Adv. Funct. Mater.* 24 (25) (2014) 3953–3961.
- [40] Z. Li, Y. Li, L. Wang, L. Cao, X. Liu, Z. Chen, et al., Assembling nitrogen and oxygen co-doped graphene quantum dots onto hierarchical carbon networks for all-solid-state flexible supercapacitors, *Electrochim. Acta* 235 (2017) 561–569.
- [41] Z. Ye, F. Wang, C. Jia, K. Mu, M. Yu, Y. Lv, et al., Nitrogen and oxygen-codoped carbon nanospheres for excellent specific capacitance and cyclic stability supercapacitor electrodes, *Chem. Eng. J.* 330 (2017) 1166–1173.
- [42] Y.-H. Lee, K.-H. Chang, C.-C. Hu, Differentiate the pseudocapacitance and double-layer capacitance contributions for nitrogen-doped reduced graphene oxide in acidic and alkaline electrolytes, *J. Power Sources* 227 (2013) 300–308.
- [43] Y.J. Oh, J.J. Yoo, Y. Kim, J.K. Yoon, H.N. Yoon, J.H. Kim, et al., Oxygen functional groups and electrochemical capacitive behavior of incompletely reduced graphene oxides as a thin-film electrode of supercapacitor, *Electrochim. Acta* 116 (2014) 118–128.
- [44] R. Ma, X. Ren, B.Y. Xia, Y. Zhou, C. Sun, Q. Liu, et al., Novel synthesis of N-doped graphene as an efficient electrocatalyst towards oxygen reduction, *Nano Res.* 9 (3) (2016) 808–819.
- [45] K. Sheng, Y. Sun, C. Li, W. Yuan, G. Shi, Ultrahigh-rate supercapacitors based on electrochemically reduced graphene oxide for ac line-filtering, *Sci. Rep.* 2 (2012) 247–251.
- [46] X. Li, B. Wei, Facile synthesis and super capacitive behavior of SWNT/MnO₂ hybrid films, *Nanomater. Energy* 1 (3) (2012) 479–487.
- [47] Z.S. Wu, A. Winter, L. Chen, Y. Sun, A. Turchanin, X. Feng, et al., Three-dimensional nitrogen and boron co-doped graphene for high-performance all-solid-state supercapacitors, *Adv. Mater.* 24 (37) (2012) 5130–5135.
- [48] B.-H. Kim, K.Y. Seung, H.-G. Woo, Boron-nitrogen functional groups on porous nanocarbon fibers for electrochemical supercapacitors, *Mater. Lett.* 93 (2013) 190–193.
- [49] H. Guo, Q. Gao, Boron and nitrogen co-doped porous carbon and its enhanced properties as supercapacitor, *J. Power Sources* 186 (2) (2009) 551–556.
- [50] Z.S. Wu, K. Parvez, X. Feng, K. Müllen, Graphene-based in-plane micro-supercapacitors with high power and energy densities, *Nat. Commun.* 4 (2013) 2487–2494.
- [51] J. Li, H. Xie, Y. Li, J. Liu, Z. Li, Electrochemical properties of graphene nanosheets/polyaniline nanofibers composites as electrode for supercapacitors, *J. Power Sources* 196 (24) (2011) 10775–10781.
- [52] D. Yu, K. Goh, H. Wang, L. Wei, W. Jiang, Q. Zhang, et al., Scalable synthesis of hierarchically structured carbon nanotube-graphene fibres for capacitive energy storage, *Nat. Nanotechnol.* 93 (7) (2014) 555–562.
- [53] S.C. Lee, U.M. Patil, S.J. Kim, S. Ahn, S.-W. Kang, S.C. Jun, All-solid-state flexible asymmetric micro supercapacitors based on cobalt hydroxide and reduced graphene oxide electrodes, *RSC Adv.* 6 (50) (2016) 43844–43854.
- [54] X. Wang, B.D. Myers, J. Yan, G. Shekhawat, V. Dravid, P.S. Lee, Manganese oxide micro-supercapacitors with ultra-high areal capacitance, *Nanoscale* 5 (10) (2013) 4119–4122.
- [55] X. Huang, H. Liu, X. Zhang, H. Jiang, High performance all-solid-state flexible micro-pseudocapacitor based on hierarchically nanostructured tungsten trioxide composite, *ACS Appl. Mater. Interfaces* 7 (50) (2015) 27845–27852.